

Fig. 5.3 Regression models and nodes discussed in this chapter

Johnson (2013) for a precise description of these nodes resp. algorithms. Moreover, some of the regression algorithms are implemented in the SPSS Modeler 18.2 to run on the SPSS Analytics Server (see IBM 2019c) and provided by the Modeler in separate nodes, like the Linear-AS node or GLE node. Since these nodes are more or less an adaptation of the nodes described in this chapter, we will ignore them here and point to IBM (2019b) for details.

5.1.2 Concept of the Modeling Process and Cross-Validation

After the data exploration and preparation steps, the actual model building process can begin. When building a model using data mining, the algorithms fits a model to a specific dataset, i.e., the training data. When fitted properly, the model describes these training data very well and has a good prediction power on it. However, a model is typically applied to new and unknown data to predict the outcome. So, the prediction ability should be nearly the same on unknown data as on the training data. This is not guaranteed. It is not uncommon that the model performance drops on new data, and thus, does not generalizes well. This phenomenon is known as *overfitting* (see section below). To validate the generalizability of the model and prevent overfitting, the build model is typically applied to unknown data to verify its universality. This process is called *cross-validation*. In particular, this method is used to evaluate classification models, such as Neural Networks or Support Vector Machines (see Chap. 8). We refer to James et al. (2013) for further information on the concept and variants of cross-validation.

The key idea behind cross-validation and creating a correct modeling process is to split data into a *training* dataset and a *test* dataset, typically in the proportion 70–30% or 80–20%. Whereas the training dataset is used to estimate the model parameters and build the model, the smaller test dataset is used to validate its performance. This gives us the chance to measure a model’s ability to generalize.

There are some important aspects to keep in mind when using a partitioning method to separate the different subsets:

1. The values in both the training and the test set should be a representation of the population, i.e., original data sample. Hence, sampling bias has to be avoided. Therefore, a random selection process should be used to separate these subsets. For sampling methods, see Sect. 2.7.8.
2. There is a trade-off between the exactness of the training results and the test results. When the data are separated into the training and the test sets, the number of records that are used for training the model is reduced. The test set, on the other hand, has to be large enough to sufficiently validate the model performance on unknown data.
3. In the area of big data, the size of the test set can be often reduced to 10% or even lower. If you have for example a set of 1 million data records, then a 10% test set has 100,000 records, which is enough to evaluate your trained model. However, you have to choose your test set large enough so that your model is properly validated!
4. The records of the test dataset cannot be used in the training procedure. The model has to be processed using only the training dataset.

Often, hyper-parameters have to be chosen prior to the model training. These parameters specify a model, but are not optimized during training. In order to find the optimal hyper-parameter, the dataset is split into three parts: the *training*, *testing*, and *validation sets*. After fitting the model for each of the parameter selections using the training data, the validation dataset is then used to find the model that will most precisely predict the target value of unknown data. This so-called “champion” model has to then be tested again with the testing dataset, to verify that it is actually suitable for use with independent data. Here, another dataset must be utilized, since the validation set is now biased, having already been used to find the optimal parameters. A typical partitioning of the dataset is training data 60%, validation data 20%, and test data 20%.

Figure 5.4 visualizes the modeling and cross-validation process. A dataset can be split into different parts with the Partition node, see Sect. 2.7.7. In Sect. 5.4.4, we will use this partitioning method by fitting a polynomial regression.

As mentioned above, cross-validation is also a very convenient way of preventing *overfitting* and is therefore often used in the modeling process for data mining tasks. For these reasons, we recommend always using a cross-validation setting when building a prediction model.

Overfitting

In statistics, overfitting occurs when a statistical model describes errors or noise instead of the underlying relationship. In other words, the model describes the training data very well, but independent and unknown data cannot be modeled. This results in poor prediction performance. Overfitting typically occurs when a model is exorbitantly complex or has too many parameters. In Fig. 5.5 overfitting in regression is illustrated. Of course, there always exists a polynomic function on which all data points are located. This is demonstrated in the left graphic. Although this is very precise for the present data, this doesn’t represent the actual structure of the data since measurement noise is modeled as well. So a linear relation describes the data structure better and is therefore a more suitable model (see the right graphic

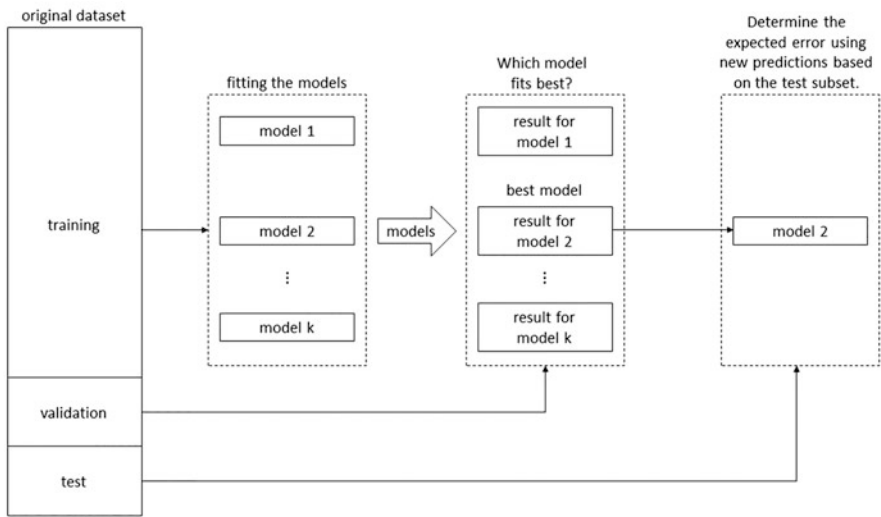


Fig. 5.4 Workflow of the modeling and cross-validation process

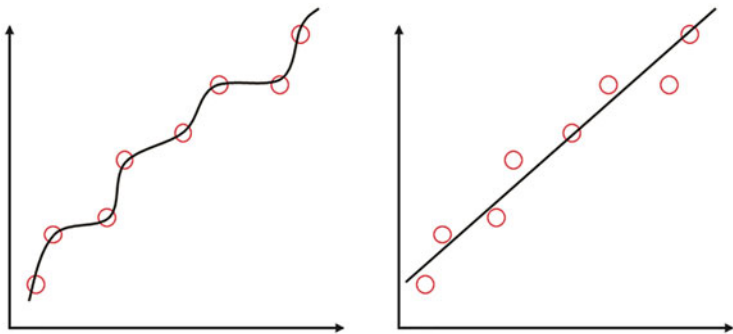


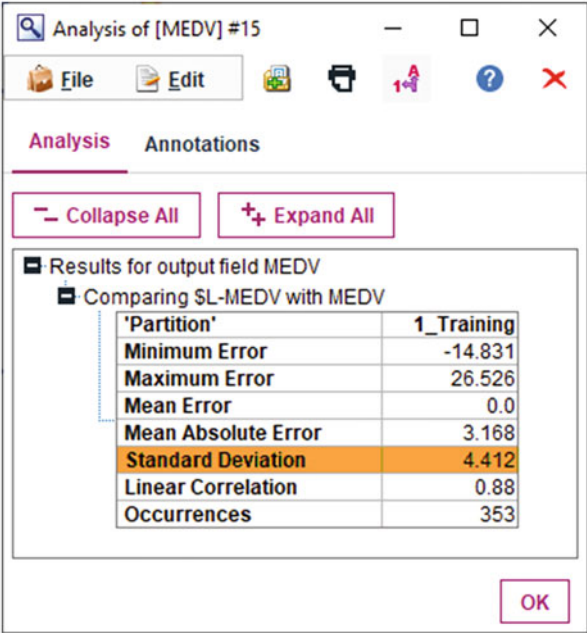
Fig. 5.5 Overfitting in regression. A complex regression function can tend to overfitting and fails to model the data structure

in Fig. 5.5). We refer to Kuhn und Johnson (2013) and James et al. (2013) for more information on overfitting.

Root Mean Squared Error: Evaluation Measure for Regression Models

A measure that indicates how well a regression model fits the data is the *root mean squared error (RMSE)*. This is just the squared sum of the residuals divided by the number of observations. In other words, it is the standard deviation of difference between the observed and predicted values (see also Sect. 5.2.1). The RMSE is a good measure for model accuracy and is used in cross-validation to compare model and their forecasting errors with each other, in order to decide which of them fits the

Fig. 5.6 Output of the Analysis node. The standard deviation gives the value of the RMSE



Analysis of [MEDV] #15

File Edit

Analysis Annotations

[-] Collapse All [+ Expand All

Results for output field MEDV

Comparing \$L-MEDV with MEDV

'Partition'	1_Training
Minimum Error	-14.831
Maximum Error	26.526
Mean Error	0.0
Mean Absolute Error	3.168
Standard Deviation	4.412
Linear Correlation	0.88
Occurrences	353

OK

data best. For more information on the RMSE, we refer to Hyndman and Koehler (2006) and Kuhn and Johnson (2013).

In a well-fitting model, the RMSE should be small. If in cross-validation, the RMSE of the testing or validation data is substantially higher than in the training data, this is an indicator of a deficient model. If, however, these values are nearly the same, the model can be used as a good predictor for independent datasets.

In the SPSS Modeler the error statistics, including the RMSE, can be calculated by the Analysis node (see Sect. 5.3.5). There the standard deviation gives us the value of the RMSE, see Fig. 5.6.

5.2 Simple Linear Regression

We start by introducing the (*simple*) *linear regression model (SLR)*, which is the easiest one in this class of regression models, since it comprises only two variables and the linear connection between them. We need to mention that a linear regression model is suitable only if a linear relationship between the two variables exists. To ensure this, we suggest performing a correlation analysis first, to verify if regression is suitable. This correlation procedure is described in Sect. 4.4.